Examiner only

9. The reversible reaction shown below was one of the first to be studied in detail in the gas and liquid phases and in solution.

$$N_2O_4 \rightleftharpoons 2NO_2$$

(a) One study of the liquid mixture showed that it contained 0.714 % NO_2 by mass. Calculate how many moles of NO_2 would be present in 25.0 g of this liquid. Give your answer to an appropriate number of significant figures. [2]

$n(NO_2)$ = ... mol

(b) Both N_2O_4 and NO_2 are soluble in a range of solvents and the equilibrium constant, K_c, can be measured in these.

(i) Give the expression for the equilibrium constant, K_c, for this equilibrium, giving its unit if any. [2]

K_c =

Unit = ...

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(ii) The equilibrium constants for the N_2O_4/NO_2 equilibrium measured at room temperature in some different solvents are listed below.

Solvent	Equilibrium constant, K_c (Unit not shown)
CS_2	17.8
CCl_4	8.05
$CHCl_3$	5.53
C_2H_5Br	4.79
C_6H_6	2.03
$C_6H_5CH_3$	1.69

I. Samples of 0.4000 mol of N_2O_4 are dissolved separately in 1 dm^3 of each of these solvents at room temperature and the reactions allowed to reach equilibrium. In one solution the concentration of N_2O_4 present at equilibrium is 5.81×10^{-2} mol dm^{-3}. Find the value of the equilibrium constant in this solution and hence identify the solvent. [3]

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Solvent ....

II. The Gibbs free energy change, ΔG, of this reaction is different in different solvents. Explain how the data shows this and state, with a reason, which solvent would have the most negative ΔG value for this reaction. [2]

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9. Cobalt forms a range of complex ions. Two of these are $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$.

(a) Draw the structures of the $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$ ions. [2]

(b) Explain why the $[Co(H_2O)_6]^{2+}$ ion is coloured. [3]

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(c) A student was given a pink-coloured solution containing $[Co(H_2O)_6]^{2+}$ ions. Upon addition of hydrochloric acid the solution turned blue as $[CoCl_4]^{2-}$ ions formed according to the equilibrium shown below.

$$[Co(H_2O)_6]^{2+} \quad + \quad 4Cl^- \quad \rightleftharpoons \quad [CoCl_4]^{2-} \quad + \quad 6H_2O$$

Aqueous silver nitrate was added to the solution containing $[CoCl_4]^{2-}$.

State and explain the observation(s) expected. [4]

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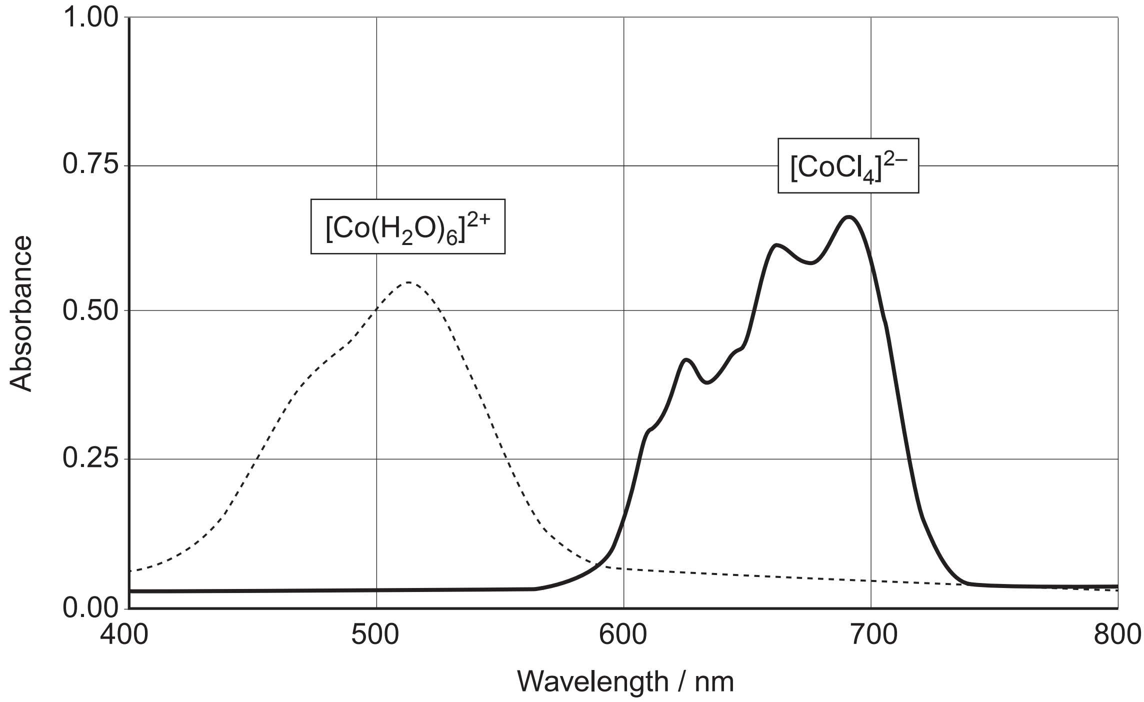
(d) These two cobalt-containing ions absorb different frequencies of light. The spectra below show the absorbance at different wavelengths of visible light by the two ions.



(i) The maximum absorbance for the $[Co(H_2O)_6]^{2+}$ ion occurs at a wavelength of 518 nm. Calculate the energy of this absorbance in kJ mol^{-1}. [3]

Energy = ... kJ mol^{-1}

(ii) A colorimeter was used to study the equilibrium between the two ions.

Suggest, giving a reason, an appropriate wavelength to use for the experiment. [1]

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(iii) Write the expression for the equilibrium constant, K_c, for this reaction **giving its unit**. [2]

$$[Co(H_2O)_6]^{2+} \quad + \quad 4Cl^- \quad \rightleftharpoons \quad [CoCl_4]^{2-} \quad + \quad 6H_2O$$

K_c =

Unit ....

(iv) A student carries out an experiment starting with separate non-aqueous solutions of $[Co(H_2O)_6]^{2+}$ and Cl^-.

When the two solutions are mixed the initial concentrations of both $[Co(H_2O)_6]^{2+}$ and Cl^- in the mixture are 0.720 mol dm^{-3}. When the mixture reaches equilibrium the concentration of the water is 0.744 mol dm^{-3}.

Calculate the value of the equilibrium constant, K_c. [3]

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K_c = ...

18

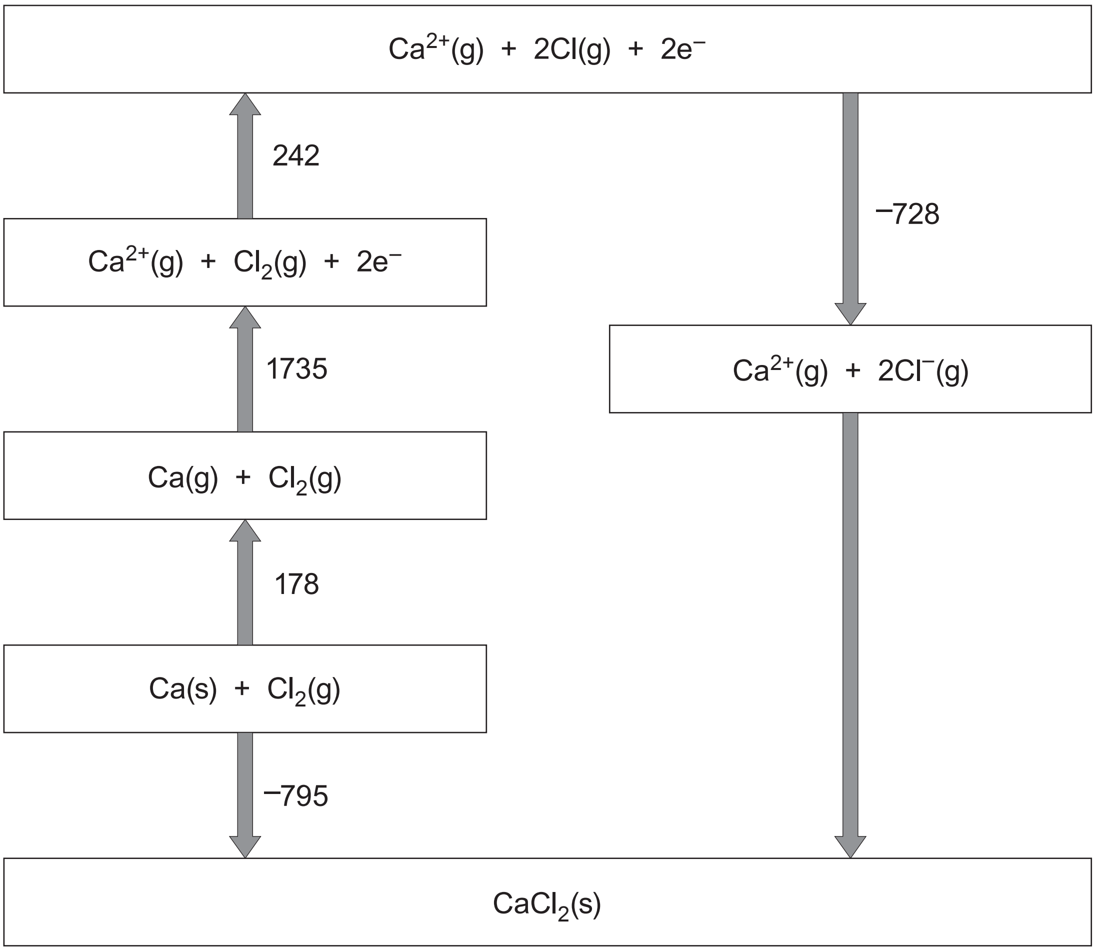
SECTION B

*Answer **all** questions in the spaces provided.*

9. Calcium chloride is an ionic solid that is used as a drying agent due to its ability to absorb water and form a range of hydrated crystals.

(a) The Born-Haber cycle below shows the formation of calcium chloride from its elements.

All values shown are standard enthalpy changes and are given in $kJ\,mol^{-1}$.



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(i) Give the value of the standard enthalpy change of atomisation of chlorine. [1]

.... $kJ\,mol^{-1}$

(ii) Calculate the standard enthalpy of lattice formation of $CaCl_2$. [2]

$\Delta_{latt\ form}H^{\theta}$ = ... $kJ\,mol^{-1}$

(iii) A student incorrectly states that the second ionisation energy of calcium is 1735 $kJ\,mol^{-1}$ as this is the energy needed to form Ca^{2+}(g) in the Born-Haber cycle.

Suggest a possible value for the second ionisation energy of calcium, giving a reason for your answer. [2]

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(b) The reaction occurring when calcium chloride is used as a drying agent is shown below.

$$CaCl_2 + xH_2O \longrightarrow CaCl_2.xH_2O$$

(i) A laboratory manual suggests a **minimum** temperature of 200 °C to remove all the water from the hydrated calcium chloride. At this temperature the reaction occurring is as follows.

$$CaCl_2.xH_2O \longrightarrow CaCl_2 + xH_2O$$

A student calculates the enthalpy change when calcium chloride becomes dehydrated as 78.0 kJ mol^{-1}.

Assuming that this data is correct, calculate a value for the entropy change when calcium chloride becomes dehydrated at this temperature. [3]

ΔS = ... J K^{-1} mol^{-1}

(ii) In the drying process 5.20 g of hydrated calcium chloride forms 3.15 g of anhydrous calcium chloride.

Calculate the value of x in the original $CaCl_2.xH_2O$. [3]

x = ...

(iii) State the colour observed in a flame test using calcium chloride. [1]

....

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(c) Calcium halides react with concentrated sulfuric acid in a similar manner to sodium halides.

State the observation(s) when concentrated sulfuric acid is added to calcium chloride and calcium bromide. Explain why the observations are different. [3]

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9. Iron is an example of a transition element. It has two main oxidation states in its compounds.

(a) Write the electronic structure of an Fe^{2+} ion and use it to show why iron is classed as a transition element. [2]

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(b) Explain why iron has more than one common oxidation state in its compounds. [1]

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(c) In very acidic solutions, Fe^{3+} forms $[Fe(H_2O)_6]^{3+}$.

Draw the structure of the $[Fe(H_2O)_6]^{3+}$ ion. [1]

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(d) In solutions with pH values between 1 and 3, the following equilibrium occurs.

$$[Fe(H_2O)_6]^{3+} \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H^+$$

The concentrations of $[Fe(H_2O)_6]^{3+}$ and $[Fe(H_2O)_5(OH)]^{2+}$ present can be studied using colorimetry.

(i) Explain why the complex ions $[Fe(H_2O)_6]^{3+}$ and $[Fe(H_2O)_5(OH)]^{2+}$ are not the same colour. [2]

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(ii) Suggest how you would select an appropriate wavelength to find the concentration of $[Fe(H_2O)_6]^{3+}$ in the equilibrium mixture. [1]

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(iii) Write an expression for the equilibrium constant K_c for this equilibrium, giving its unit. [2]

Unit ....

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(iv) A known mass of iron(III) chloride, $FeCl_3.6H_2O$ (M_r 270.4) is dissolved in 1.00 dm^3 of dilute hydrochloric acid. At equilibrium the pH of the solution is 1.55 and the concentration of $[Fe(H_2O)_5(OH)]^{2+}$ is 0.103 mol dm^{-3}. The numerical value of K_c under these conditions is 4.03×10^{-3}.

Find the mass of $FeCl_3.6H_2O$ used to make the solution. [4]

Mass = ... g

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(e) At higher pH values, solutions containing hydrated Fe^{3+} ions form a precipitate.

(i) Give the formula of the precipitate and its colour. [1]

Formula ....

Colour ....

(ii) Aqueous iron(II) compounds form a precipitate of a different colour with aqueous sodium hydroxide. In a test tube, the precipitate can change colour on standing. Use the standard electrode potential values below to explain this change. [2]

	Standard electrode potential, E^{θ} / V
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77
$O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$	+1.23

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(iii) A student states that it would be incorrect to use the standard electrode potential for the oxygen half-cell as the solution will be alkaline. State, giving a reason, the effect an alkaline solution will have on the value of the electrode potential of the oxygen half-cell. [2]

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(f) In a blast furnace Fe_2O_3 is reduced by CO to form iron metal.

(i) Write an equation for this process. [1]

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(ii) Explain why CO is a good reducing agent. [1]

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Turn over.

Examiner only

9. (a) The elements of the p-block show a variety of oxidation states within the same group, whilst some elements of the d-block can show an even wider range of oxidation states.

- Some oxides of group 4 have an oxidation state of +4 such as CO_2, SiO_2 and PbO_2 and others have an oxidation state of +2 such as CO and PbO. There is no stable oxide with the formula SiO.
- Some chlorides of group 5 have a +5 oxidation state such as PCl_5 and some have an oxidation state of +3 such as NCl_3 and PCl_3. There is no stable chloride with the formula NCl_5.
- Manganese forms a range of oxides including MnO, Mn_2O_3 and MnO_2 which have oxidation states of +2, +3 and +4 respectively.

Explain why these elements can form the oxidation states shown and why SiO and NCl_5 do not form. [6 QER]

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(b) Tin(II) ions, Sn^{2+} (aq), can be used as reducing agents for a range of metal ions. In one study of the reduction of Pt^{4+} ions to Pt^{2+} ions in an acid solution, the reaction was found to be first order with respect to both Pt^{4+} and Sn^{2+} with a rate constant of $895\,mol^{-1}\,dm^3\,s^{-1}$ at 297 K.

(i) Write a rate equation for this reaction. [1]

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(ii) A student suggests studying this reaction using sampling and quenching with samples taken every 30 seconds for 5 minutes. Explain why this would not be an appropriate sampling interval. [2]

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(c) (i) A second student decides to use colorimetry to study the rate of this reaction. Explain what the student should consider when choosing an appropriate wavelength of light for their colorimetry experiment. [1]

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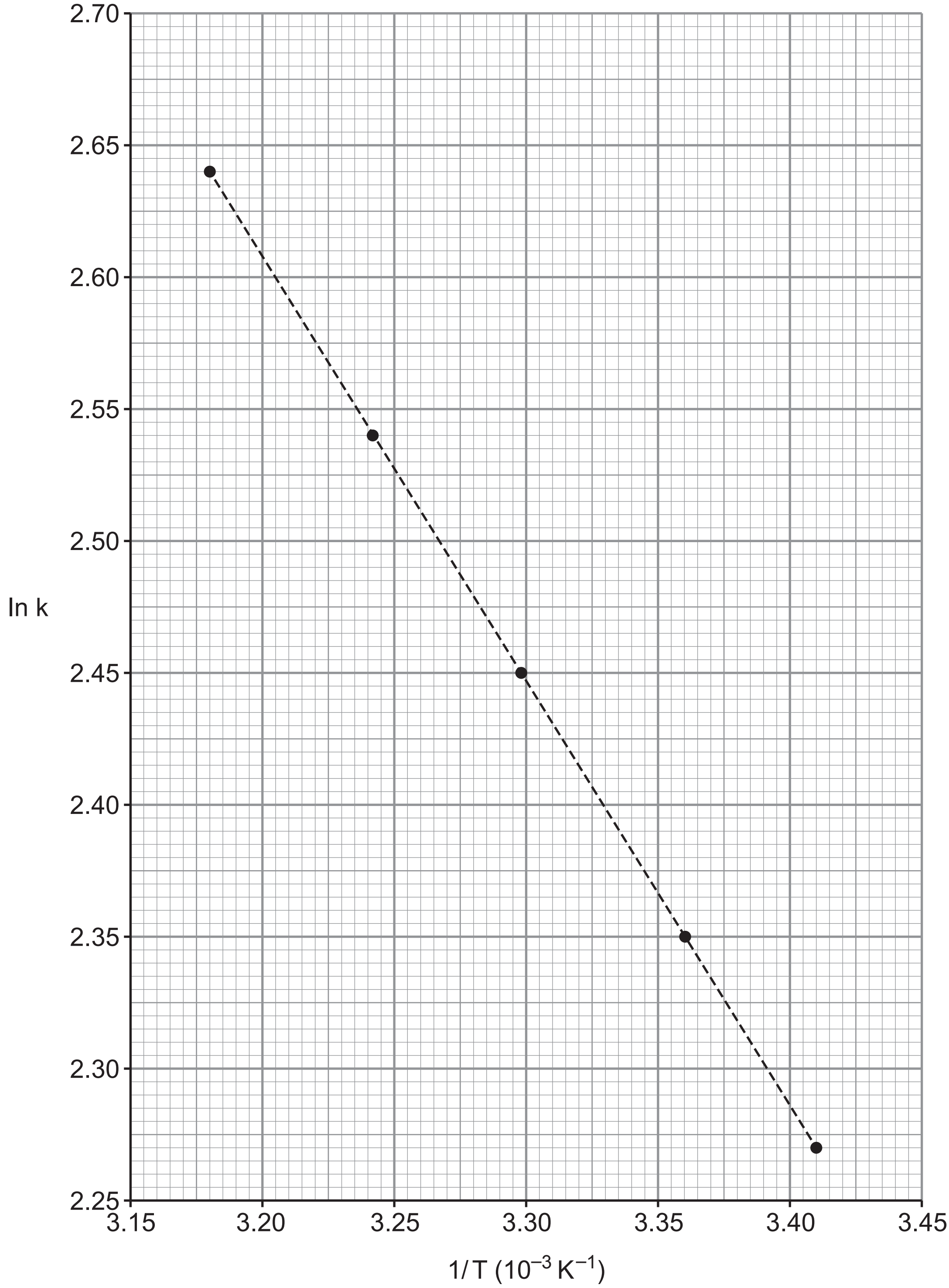
(ii) The value of the rate constant was measured at five different temperatures to attempt to find the activation energy.

$$\ln k = \ln A - (E_a/RT)$$

A graph was plotted of (ln k) against (1/Temperature) and this is shown below.



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Find the activation energy of the reaction. [3]

activation energy, E_a = ... $kJ\,mol^{-1}$

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